

between the screen and the projector, it is possible to increase the video size without significant additional expenditure. The projector used can be of the CRT, LCD, or DLP type.

4.1.22 Refresh Rate

In a CRT display, the refresh rate is the number of times per second that the electron beam scans through the screen. Since the beam travels row-wise from the top to the bottom, CRT displays have two refresh rates associated. The Horizontal Refresh Rate is the number of rows scanned by the beam per second. This is usually in the order of Kilohertz and is of not much importance. The Vertical Refresh Rate is the number of times all the rows in the screen are refreshed, and this is in the range of 50 to 150 Hertz. Normal references to “refresh rate” are to the Vertical Refresh Rate.

4.1.23 Resolution

The total number of pixels that make up a display is its resolution; it is normally denoted as the number of pixels on the horizontal axis and on the vertical axis. A screen with 1024 x 768 resolution has 768 rows of pixels with each row having 1024 pixels.

4.1.24 Response time

In LCDs, since the source of light is always on, the concept of refreshing is not applicable. Response time refers to the time it takes the liquid crystal to change shape. Since changing the shape also changes the light passing to the screen, response time is defined in two alternative ways. Black-to-black response time refers to the time it takes the liquid crystal to switch from a shape that allows no light to pass (black) to a shape that allows all light to pass (white) and revert. Grey-to-grey response time refers to the time it takes the liquid crystal to switch from any arbitrary shape to any other arbitrary shape, making a pixel change colour from one shade of grey to another. There is no common pair of shades of grey that is used by all manufacturers. Response time is usually mentioned in milliseconds.

Response time is an important factor that influences the display's performance when dealing with fast-moving images. If the